

Calibration as Decision Alignment

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- ▶ Calibration is a self-consistency requirement: forecasts must be unbiased conditional on their own value.
- ▶ It makes taking the forecast at face value at least as good as any alternative rule using only the forecast.

Forecasts Are Decision Interfaces

In the first part of the course, an alignment procedure produced the policy whose welfare we evaluated:

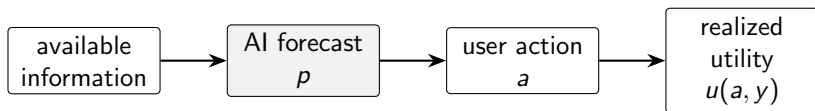
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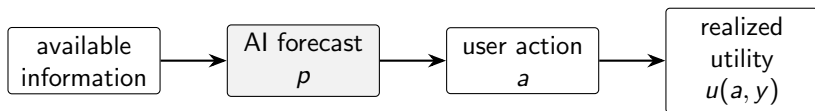


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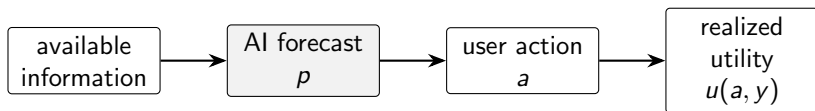
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What property lets downstream decision makers safely treat forecasts as if they are correct?

The User Maximizes Expected Utility Under the Forecast

Let $\mathcal{Y} = \{1, \dots, d\}$ be a finite outcome set and let $p \in \Delta(\mathcal{Y})$ be a forecast. A user has a finite action set $\mathcal{A}$ and utility $u(a, y)$.

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Represent action a by the vector

$$u_a = (u(a, 1), \dots, u(a, d)) \in \mathbb{R}^d.$$

Its expected utility under outcome distribution p is

$$\mathbb{E}_{Y \sim p}[u(a, Y)] = \sum_{y \in \mathcal{Y}} p_y u(a, y) = \langle u_a, p \rangle.$$

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Fixing a tie-breaking rule, the user's best response action is

$$\text{BR}_u(p) \in \arg \max_{a \in \mathcal{A}} \langle u_a, p \rangle.$$

Running Example: Deploy, Hold, or Audit

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Let r denote the reported probability that the proposal is safe, so the forecast vector is $p = (r, 1 - r)$.

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Write $BR_u(r)$ as shorthand for $BR_u((r, 1 - r))$.

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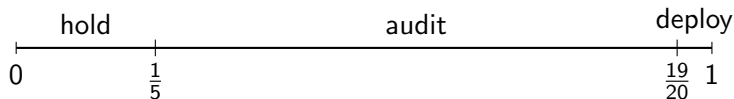
$$U(\text{audit}; r) = r\left(\frac{4}{5}\right) + (1 - r)\left(-\frac{1}{5}\right) = r - \frac{1}{5}.$$

Therefore, away from the two ties,

$$BR_u(r) = \begin{cases} \text{hold}, & r < 1/5, \\ \text{audit}, & 1/5 < r < 19/20, \\ \text{deploy}, & r > 19/20. \end{cases}$$

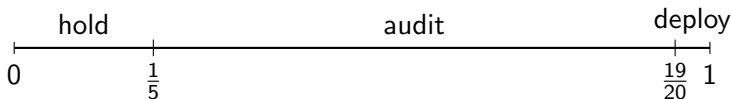
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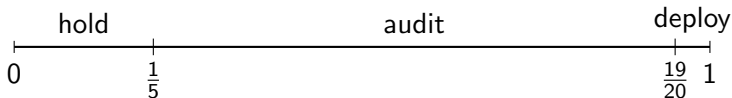
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Changing a forecast from 0.19 to 0.21 does change the action.

Errors near a decision boundary can therefore matter more than equally large errors elsewhere.

Unbiased Forecasting On Average Is Not Enough

Consider equally many rounds of two kinds:

reported safe probability	outcome	chosen action	utility
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Yet the user deploys every harmful proposal and holds every safe proposal. The errors cancel only because we averaged across reports that induce different decisions.

Calibration Conditions on What Was Reported

We work with a fixed realized sequence

$$(p_1, y_1), \dots, (p_T, y_T), \quad p_t \in \mathcal{P} \subset \Delta(\mathcal{Y}), \quad y_t \in \mathcal{Y},$$

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Let $N_p := |\{t : p_t = p\}|$. For a report with $N_p > 0$, let e_y denote the point mass on outcome y and define the empirical outcome distribution in that report cell:

$$q_p := \frac{1}{N_p} \sum_{t: p_t = p} e_{y_t}.$$

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The sequence is exactly calibrated when $q_p = p$ for every report that occurs. For binary outcomes: among rounds assigned probability r , the positive outcome occurs with frequency r .

Report-Cell Bias Measures Approximate Calibration

It is convenient to avoid dividing by the size of a report cell.
Define its unnormalized bias

$$B_p := \sum_{t:p_t=p} (e_{y_t} - p).$$

When $N_p > 0$, this is equivalently $B_p = N_p(q_p - p)$. Exact calibration means $B_p = 0$ for every $p \in \mathcal{P}$.

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Population analogue: if $\hat{P}$ is a random forecast, calibration requires

$$\mathbb{E}[e_Y \mid \hat{P} = p] = p \quad \text{for every } p \text{ in the support of } \hat{P}.$$

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Its average realized gain over the best response policy is

$$R_T^{\text{rep}}(\pi) := \frac{1}{T} \sum_{t=1}^T (u(\pi(p_t), y_t) - u(a_t, y_t)).$$

Is reinterpretation ever worthwhile?

First Group the Realized Gain by Report

Write $a(p) = \text{BR}_u(p)$. Since $u(a, y) = \langle u_a, e_y \rangle$,

$$\begin{aligned} TR_T^{\text{rep}}(\pi) &= \sum_{t=1}^T \langle u_{\pi(p_t)} - u_{a(p_t)}, e_{y_t} \rangle \\ &= \sum_{p \in \mathcal{P}} \left\langle u_{\pi(p)} - u_{a(p)}, \sum_{t: p_t=p} e_{y_t} \right\rangle. \end{aligned}$$

The definition of B_p gives

$$\sum_{t: p_t=p} e_{y_t} = N_p p + B_p.$$

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Substituting $\sum_{t:p_t=p} e_{y_t} = N_p p + B_p$ gives

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The first sum is nonpositive because $a(p) = \text{BR}_u(p)$ maximizes $\langle u_a, p \rangle$.

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Define the largest coordinatewise utility difference between two actions:

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Therefore

$$TR_T^{\text{rep}}(\pi) \leq D_u \sum_{p \in \mathcal{P}} \|B_p\|_1.$$

Calibration Rules Out Profitable Reinterpretation

Theorem

For every fixed forecast and outcome sequence and every reinterpretation $\pi : \mathcal{P} \rightarrow \mathcal{A}$,

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Because full calibration does not mention u , the conclusion holds for every downstream utility function and action set.

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Now suppose an additional feature x_t reveals whether the proposal is safe. A contextual rule that sees x_t deploys exactly on safe rounds and holds on harmful rounds, obtaining average utility $1/2$. There is no contradiction: the theorem compares only with reinterpretations of the constant report. Calibration cannot recover information not present in the forecast.

Errors Can Cancel Within One Decision Region

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The prediction conditional bias cancels after the two report cells are combined. That cancellation is harmless for a user who takes the same action on both cells.

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Its average gain is

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The user's average swap regret is

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A small value means that hindsight reveals no systematic criticism of the form

Whenever the forecast led me to action a , I should instead have taken $\phi(a)$.

Constant remappings $\phi(a) \equiv b$ are included, so this benchmark is only stronger than the fixed-action comparisons measured by external regret in Lecture 5.

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This condition allows errors from distinct reports to cancel—but only when those reports induced the same action for this user. For any nonempty action cell, this says that the average forecast equals the empirical outcome distribution within the rounds inducing that action.

Full Calibration Implies Decision Calibration

Because the action is a deterministic function of the report,

$$\{t : a_t = a\} = \bigcup_{p: \text{BR}_u(p)=a} \{t : p_t = p\}.$$

The report cells in this union are disjoint, so

$$\begin{aligned} B_a &= \sum_{t: a_t=a} (e_{y_t} - p_t) \\ &= \sum_{p: \text{BR}_u(p)=a} \sum_{t: p_t=p} (e_{y_t} - p) \\ &= \sum_{p: \text{BR}_u(p)=a} B_p. \end{aligned}$$

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Thus $B_p = 0$ for every report implies $B_a = 0$ for every action. The 0.30/0.50 example shows that the converse fails.

The Same Transfer Argument Now Groups by Action

Fix an action remapping ϕ . Its total realized gain is

$$TR_T(\phi) = \sum_{t=1}^T \langle u_{\phi(a_t)} - u_{a_t}, e_{y_t} \rangle$$

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Write $e_{y_t} = p_t + (e_{y_t} - p_t)$ to separate the forecasted comparison from the forecast error:

$$\begin{aligned} TR_T(\phi) = & \underbrace{\sum_{t=1}^T \langle u_{\phi(a_t)} - u_{a_t}, p_t \rangle}_{\text{comparison in the forecasted world}} \\ & + \underbrace{\sum_{t=1}^T \langle u_{\phi(a_t)} - u_{a_t}, e_{y_t} - p_t \rangle}_{\text{cost of replacing forecasts by outcomes}} . \end{aligned}$$

Because $a_t = \text{BR}_u(p_t)$, every term in the first sum is nonpositive.

Action-Cell Bias Controls the Remaining Error

On every round in the action- a cell, the utility difference $u_{\phi(a_t)} - u_{a_t}$ equals the fixed vector $u_{\phi(a)} - u_a$. Therefore

$$\begin{aligned} & \sum_{t=1}^T \langle u_{\phi(a_t)} - u_{a_t}, e_{y_t} - p_t \rangle \\ &= \sum_{a \in \mathcal{A}} \left\langle u_{\phi(a)} - u_a, \sum_{t: a_t = a} (e_{y_t} - p_t) \right\rangle \\ &= \sum_{a \in \mathcal{A}} \langle u_{\phi(a)} - u_a, B_a \rangle. \end{aligned}$$

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Recall that

$$D_u = \max_{a, b \in \mathcal{A}} \|u_a - u_b\|_{\infty}.$$

For each action cell, Hölder's inequality gives

$$|\langle u_{\phi(a)} - u_a, B_a \rangle| \leq \|u_{\phi(a)} - u_a\|_{\infty} \|B_a\|_1 \leq D_u \|B_a\|_1.$$

Decision Calibration Controls Swap Regret

Theorem

For every fixed forecast and outcome sequence,

$$\text{SwapReg}_T \leq \frac{D_u}{T} \sum_{a \in \mathcal{A}} \|B_a\|_1.$$

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$$\text{SwapReg}_T \leq \frac{D_u}{T} \sum_{a \in \mathcal{A}} \|B_a\|_1.$$

Exact decision calibration therefore gives

$$\text{SwapReg}_T = 0.$$

The forecast need not be calibrated separately at every reported probability; it needs its residual errors to cancel on the cells relevant to the user's decision function

Calibration Types and Their Benchmarks

condition	errors grouped by	Benchmark class controlled
full calibration	reported forecast p	every reinterpretation $\pi(p)$
decision calibration	induced action $a = \text{BR}_u(p)$	every action remapping $\phi(a)$

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Full calibration is utility-independent and serves arbitrary users. Decision calibration is weaker, and its cells depend on the specified users' utilities and action sets. We will see decision calibration is easier to obtain in high dimensional settings.

Decision-Calibrated Forecasting Appears Circular

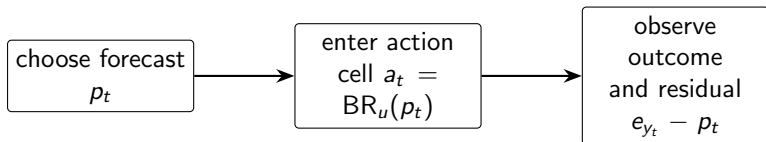
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Decision-Calibrated Forecasting Appears Circular

So far we have assumed small action-cell biases. Can a forecaster keep them small even in adversarial sequential forecasting settings? For action a , the active cell is the event

$$E_a(p_t) := \mathbf{1}\{\text{BR}_u(p_t) = a\}.$$

Choosing p_t determines which cell is active; only afterward does the outcome reveal the residual that updates its bias:



Next, we will see that Blackwell approachability extends Lecture 5's minimax argument to vector payoffs, choosing forecasts whose joint action-cell bias approaches a small ball around zero.

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Summary

- ▶ A probability forecast provides a decision interface: the user chooses an action maximizing expected utility under the report.
- ▶ Full calibration makes residual errors cancel within every report cell, ruling out profitable reinterpretations of the report.
- ▶ Calibration guarantees that it is optimal to “trust the forecast and act accordingly” amongst all forecast-based policies.
- ▶ For a fixed user, action-cell calibration is enough to control swap regret and may be strictly weaker than full calibration. Hopefully it is easier to achieve.

Thanks!

See you next class!

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